Qn. Set Code-7(SOT)

Semester: 6th
Programme: B.Tech
Branch: CSE, IT, CSSE

SPRING END SEMESTER EXAMINATION-2025

6th Semester B.Tech

ARTIFICIAL INTELLIGENCE

CS30002 / CS3011

(For 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Differentiate between rational agent and omniscient agent.

(b) What is the basic difference between breadth first search and uniform cost search algorithm?

(c) Define absolute independence and conditional independence in probability theory.

(d) First-order logic (FOL) is superior to Prepositional logic (True/False). Justify with reason.

(e) What is a knowledge-based agent?

(f) Given 3 literals a, b, and c, using this how many possible models may exist for the statement: $a \vee \neg b \vee c$.

(g) List four performance measures for Uniform Cost Search algorithm.

(h) Alpha – Beta pruning is better than mini-max algorithm. True/False. Justify with season.

(i) Mention about the term shoulder, platue in hill-climbing algorithm.

(j) What is Bayesian Network? Give an example.

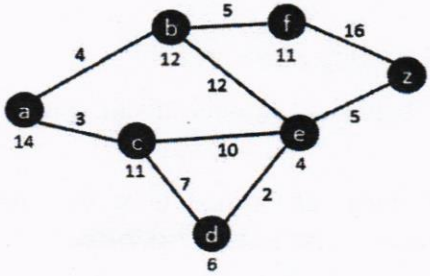
SECTION-B

2. (a) Explain Task environment, State space and PEAS representation. Give the PEAS representation for Vacuum-Cleaner world along with its State space diagram and write its Agent function. Give some examples of the percept sequence and action mapping of this agent. [5]

(b) Find the order of expansion of nodes for the following search graph by applying [5]

i) Uniform cost search

ii) A* search



Assume start node is a and goal node is z. The value indicated in figure having proper meaning (z having heuristic value is zero).

3. (a) Mention different types of agent. Explain Utility-based agent by considering suitable example along with proper diagram. [5]

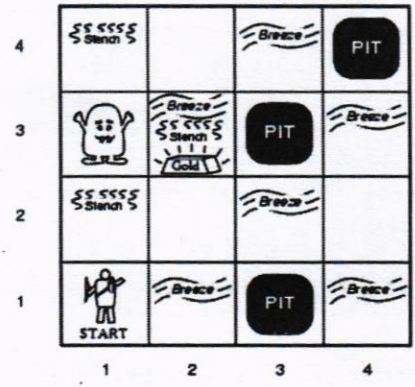
(b) Let's think about a situation, there are 3 jugs of capacities 12, 8 and 5 liters respectively. There is no scale on the jugs. So it is only their capacities which are only known. Initially, the 12 liter jug is full of water [5]

and the other two jugs are empty. The water can be poured from one jug to another. It is important to note that during the process neither any extra water is added from any outside source nor any water is thrown out permanently from any jug to outside. The goal is to have exactly 6 liters of water in each of any two jugs.

Find the step-by-step solution of this problem by making a proper representation of each state.

4. (a) Write the difference between Informed Search and Uninformed Search. You are given a 2D grid with obstacles and a start and end point. Which search algorithm would you choose to find the shortest path and why? [5]

(b) What is Logical Agent? What are its properties? Demonstrate the Wumpus world problem and its solution with the help of logical agents for given scenario and proof Wumpus present in [1, 3] location (all symbols in figure have proper meaning). [5]

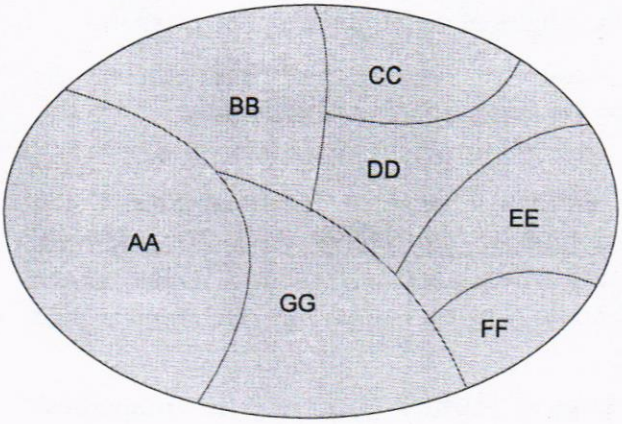


5. (a) Define the following Cryptarithmetic problem as a CSP: [5]

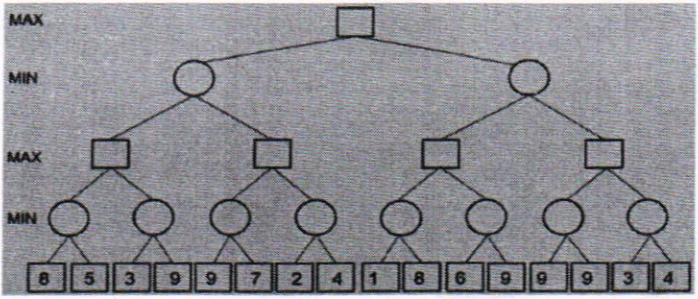
CKIBB + KIGFB = FGHIJK

Solve this problem and find the values for all symbols used. (Apply constraint satisfaction rules.)

(b) Formally define map coloring problem as constraint satisfaction problems (CSP). Solve map coloring problem of the following graph using constraint propagation and forward checking process. [5]



6. (a) Explain α-β pruning and how it works. Also solve the following example and show how α-β pruning helps in this scenario. [5]



(b) (i) Differentiate between Propositional Logic and First Order Logic. [2 + 3]

(ii) What is PDDL in classical planning? Determine PDDL descriptions in case of Air Cargo Problem and the Blocks World?

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